BANK MANAGEMENT SYSTEM USING JAVA

1. ABSTRACT

The Bank Management System is a simple Java-based mini project developed to perform basic banking operations. The main purpose of this project is to make banking activities simple and easy to understand.

This system allows users to enter customer details, deposit money, withdraw money, check account balance, and display account details. It uses basic Java programming concepts such as variables, methods, conditional statements, switch statements, and loops.

The project provides a menu-driven interface that allows users to select different banking operations. It helps beginners understand how Java programming can be used to develop real-world applications.

2. INTRODUCTION

- Bank Management System is a software application designed to manage basic banking activities. In traditional banking systems, maintaining customer details and calculating account balances manually can take more time.
- This project provides a simple computerized solution for basic banking operations using Java. The user can enter customer details and perform deposit, withdrawal, and balance enquiry operations.
- The project is developed using Core Java and provides an easy-to-use console interface.

2.1 Problem Statement

To develop a simple Java application that performs basic banking operations such as account details management, depositing money, withdrawing money, and checking account balance.

2.2 Objectives

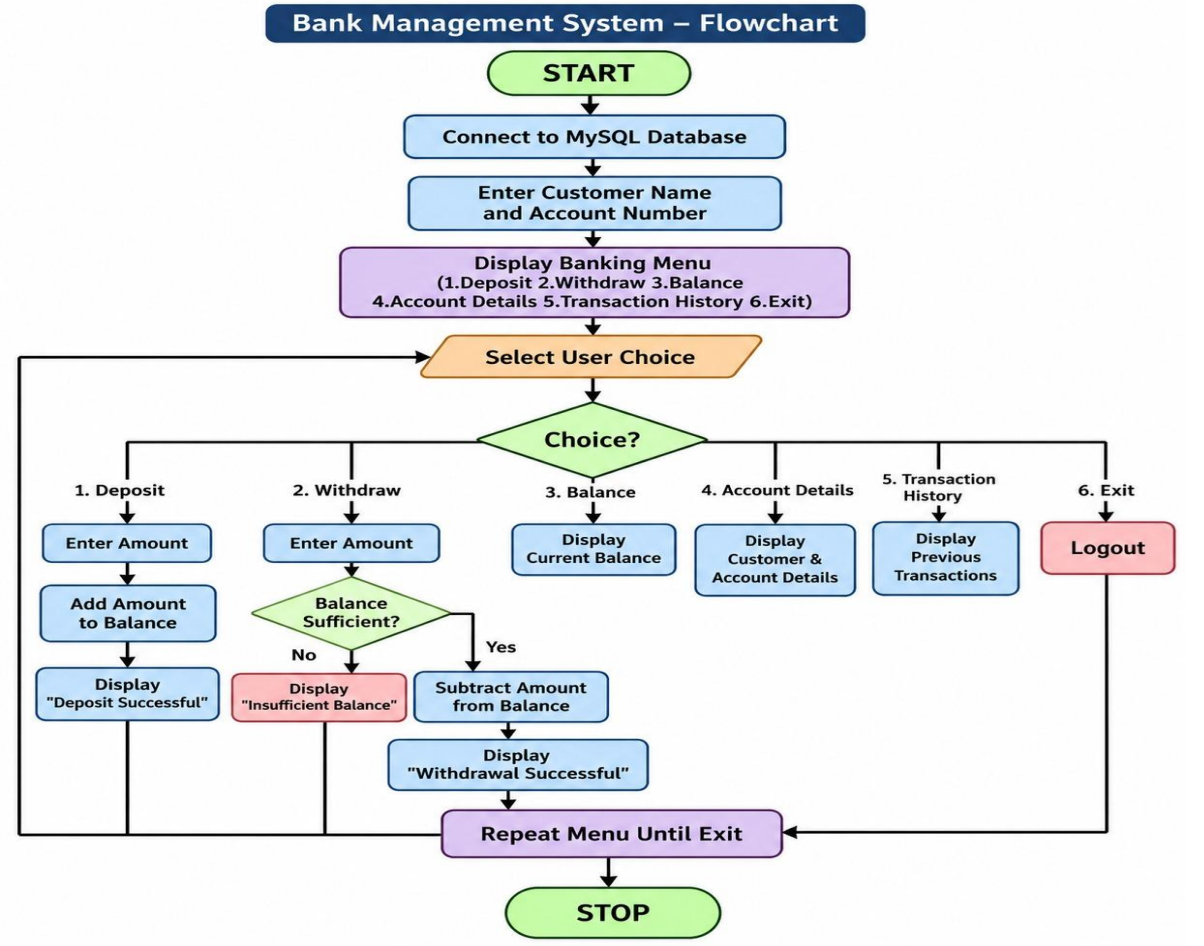
- To develop a simple banking application using Java.
- To display customer account details.
- To perform deposit operations.

- To perform withdrawal operations.
- To check the available account balance.
- To understand basic Java programming concepts.

3. ALGORITHM

- Start the program.
- Import the Scanner package.
- Read the customer name.
- Enter the account number.
- Display the banking menu.
- Ask the user to select an option.
- If the user selects Deposit, enter the amount and add it to the balance.
- If the user selects Withdraw, check the available balance and subtract the amount.
- If the user selects Check Balance, display the current balance.
- If the user selects Account Details, display customer information.
- If the user selects Exit, terminate the program.
- Repeat the menu until the user selects Exit.
- Stop the program.

4. FLOWCHART



5. METHODOLOGY

The methodology explains the step-by-step process used to develop the Bank Management System.

5.1 Requirement Analysis

The system requires the following hardware and software:

Hardware Requirements:

- Computer or Laptop
- Minimum 2 GB RAM
- Keyboard and Monitor

Software Requirements:

- Java Development Kit (JDK)

- Visual Studio Code / Eclipse / IntelliJ IDEA
- Operating System: Windows or Linux

5.2 System Design

The system is designed using a menu-driven approach. The user enters customer details and selects the required banking operation.

The system contains the following main components:

- Customer Details
- Account Number
- Deposit Operation
- Withdrawal Operation
- Balance Enquiry
- Account Details Display

5.3 Implementation

The project is implemented using Java programming language.

The Scanner class is used to accept user input. Variables are used to store customer information and account balance. Switch-case statements are used to select banking operations, and a do-while loop repeatedly displays the menu.

5.4 Input Module

The input module accepts the following information from the user:

- Customer Name
- Account Number
- Menu Choice
- Deposit Amount
- Withdrawal Amount

5.5 Processing Module

The processing module performs the selected banking operations.

- Deposit: Adds the entered amount to the account balance.

- Withdrawal: Subtracts the amount from the balance if sufficient funds are available.
- Balance Enquiry: Retrieves the current account balance.
- Account Details: Displays customer information.

5.6 Decision Module

- The decision module uses if-else and switch-case statements.
- It checks the user's selected option and determines which banking operation should be performed.
- It also checks whether the withdrawal amount is valid and whether sufficient balance is available.

5.7 Looping Module

- The do-while loop is used to display the banking menu repeatedly.
- The menu continues until the user selects the Exit option.

5.8 Output Module

The output module displays the results of the banking operations.

The system displays:

- Account details
- Deposit success message
- Withdrawal success message
- Available account balance
- Invalid choice message
- Exit message

RESULT

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================================
     BANK MANAGEMENT SYSTEM
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1. Customer
2. Admin
3. Exit
Enter choice: 1

--- CUSTOMER MENU ---
1. Register
2. Login
3. Back
Enter choice: 2
Username: Boddeti Nagasatya
Password: Satya123
Invalid username or password.

--- CUSTOMER MENU ---
1. Register
2. Login
3. Back
Enter choice:
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```
--- CUSTOMER MENU ---
1. Register
2. Login
3. Back
Enter choice: 1
Create username: Boddeti Nagasatya
Create password: Satya123
Registration successful!

--- CUSTOMER MENU ---
1. Register
2. Login
3. Back
Enter choice: 2
Username: Boddeti Nagasatya
Password: Satya123
Login successful!
```

```
--- CUSTOMER MENU ---
1. Register
2. Login
3. Back
Enter choice: 1
Create username: Boddeti Nagasatya
Create password: Satya123
Registration successful!

--- CUSTOMER MENU ---
1. Register
2. Login
3. Back
Enter choice: 2
Username: Boddeti Nagasatya
Password: Satya123
Login successful!
```

```
--- CUSTOMER DASHBOARD ---
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. View Transactions
6. Logout
Enter choice: 3
Enter account number: 1001
Enter withdrawal amount: 200
Withdrawal successful!
Remaining Balance: Rs. 300

--- CUSTOMER DASHBOARD ---
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. View Transactions
6. Logout
Enter choice:

=== Execution Timed Out ===
```

Run

7. CONCLUSION

- The Bank Management System was successfully developed using Java programming language. The project performs basic banking operations such as depositing money, withdrawing money, checking account balance, and displaying customer details.
- This project helped in understanding important Java concepts such as variables, Scanner class, switch-case statements, if-else conditions, and do-while loops.
- The system is simple, user-friendly, and useful for beginners who want to learn Java application development.

8. FUTURE ENHANCEMENTS

The project can be improved by adding more advanced features in the future.

- Customer Registration: Add complete customer registration with personal details, mobile number, email address, and customer ID.
- ATM / Debit Card System: Introduce a simulated ATM or debit card facility for balance enquiry and demo transactions.
- Discount Offers: Provide sample banking offers and cashback-style rewards for customers.
- Database Connectivity: Integrate MySQL to store customer information and transaction details permanently.
- Login System: Add username and password authentication for customer accounts.
- Graphical User Interface: Develop an attractive interface using Java Swing or JavaFX.